

DONGQI LI

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Education

University of Kentucky

Ph.D. in Physics, expected 2029

Advisor: Chunli Huang.

Lexington, KY, USA

Aug. 2024 – Present

Tsinghua University

Visiting Student, Yau Mathematical Sciences Center

Advisor: Ling-Yan Hung.

Beijing, China

Sep. 2023 – Jun. 2024

Sun Yat-sen University

M.S. in Theoretical Physics

Advisor: Rong-Xin Miao.

Zhuhai, China

Sep. 2020 – Jun. 2023

Thesis: Holographic Research on Black Hole Information Paradox.

Sun Yat-sen University

B.S. in Physics

Advisor: Rong-Xin Miao.

Zhuhai, China

Aug. 2016 – Jun. 2020

Thesis: Entanglement Entropy Growth During Holographic Thermalization.

Research Experience

Neural Quantum States for Anyons

Self-attention; Jastrow transformation; variational Monte Carlo

University of Kentucky

2025 – Present

- Develop neural-network variational Monte Carlo methods for Abelian anyons in two-dimensional harmonic traps.
- Construct symmetry-preserving self-attention wave functions incorporating anyonic short-distance structure and particle-exchange symmetry, with few-body benchmarks against established numerical results.

Quantum Monte Carlo for Spin-Valley Ferromagnetism

SSE sampling; finite-size scaling; helicity modulus

University of Kentucky

2024 – Present

- Study finite-temperature spin-valley phases in rhombohedral multilayer graphene using stochastic-series-expansion quantum Monte Carlo.
- Characterize the BKT transition through finite-size scaling and helicity-modulus analysis, including the effects of spin-orbit coupling, anisotropy, and external fields.

Tensor Networks and Conformal Field Theory

Category theory; fusion rules; tensor networks

Tsinghua University

Sep. 2023 – Jun. 2024

- Studied critical properties of (2+1)-dimensional conformal field theories using tensor-network methods based on category-theoretic structures and fusion rules.

Holography and Black-Hole Information

Sun Yat-sen University

RT-surface numerics; higher-curvature gravity; Weyl anomaly

2019 – 2023

- Studied black-hole information, holographic entanglement entropy, and anomaly-induced currents.

Publications

1. **D. Li** and R.-X. Miao, “Massless Entanglement Islands in Cone Holography,” *JHEP* **06** (2023) 056. [doi:10.1007/JHEP06\(2023\)056](https://doi.org/10.1007/JHEP06(2023)056); [arXiv:2303.10958](https://arxiv.org/abs/2303.10958).
2. P.-J. Hu, **D. Li**, and R.-X. Miao, “Island on codimension-two branes in AdS/dCFT,” *JHEP* **11** (2022) 008. [doi:10.1007/JHEP11\(2022\)008](https://doi.org/10.1007/JHEP11(2022)008); [arXiv:2208.11982](https://arxiv.org/abs/2208.11982).
3. Q.-L. Hu, **D. Li**, R.-X. Miao, and Y.-Q. Zeng, “AdS/BCFT and Island for curvature-squared gravity,” *JHEP* **09** (2022) 037. [doi:10.1007/JHEP09\(2022\)037](https://doi.org/10.1007/JHEP09(2022)037); [arXiv:2202.03304](https://arxiv.org/abs/2202.03304).
4. J.-J. Zheng, **D. Li**, Y.-Q. Zeng, and R.-X. Miao, “Anomalous current due to Weyl anomaly for conformal field theory,” *Phys. Lett. B* **797** (2019) 134844. [doi:10.1016/j.physletb.2019.134844](https://doi.org/10.1016/j.physletb.2019.134844); [arXiv:1904.07017](https://arxiv.org/abs/1904.07017).

Presentations and Posters

- **Oral presentation:** D. Li and C. Huang, “[Finite temperature properties of half-metals in rhombohedral multilayer graphene](#),” APS Global Physics Summit, Anaheim, CA, Mar. 18, 2025.
- **Poster:** “Massless entanglement islands in cone holography,” First International Congress of Basic Science (ICBS), Beijing, China, Jul. 16, 2023.
- **Oral presentation:** “Page curves in codim-n holography,” Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society, Chongqing, Apr. 23, 2023.
- **Poster:** “Novel phase transition of Page curve for Gauss-Bonnet gravity,” 1st SUIAS Workshop in HEP: Supersymmetry and Gravitation, online, Aug. 2022.
- **Oral presentation:** “Novel phase transition of Page curve for Gauss-Bonnet gravity,” Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society, Liyang, China, Jul. 4, 2022.
- **Poster:** “Novel phase transition of Page curve for Gauss-Bonnet gravity,” 23rd International Conference on General Relativity and Gravitation (GR23), online, Jul. 3, 2022.

Honors

National Scholarship (China), 2022.

Teaching

University of Kentucky

2025 – Present

PHY 211 recitation; PHY 241/242 laboratories; PHY 361/228 grading.

Sun Yat-sen University

2021 – 2022

Teaching Assistant, Electrodynamics and Electromagnetism.